Practical 6: Clustering Algorithms - Submission Template

Course: 316316 - Machine Learning | **Practical:** 6 of 15 | **Status:** Complete

Student Information

Field	Value
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Learning Outcomes - All 10 Achieved ✅

LO	Outcome	Status
1	Clustering concepts	☑ Yes
2	K-Means	☑ Yes
3	Hierarchical	☑ Yes
4	DBSCAN	☑ Yes
5	Optimal clusters	☑ Yes
6	Silhouette score	☑ Yes
7	Visualizations	☑ Yes
8	Compare algorithms	☑ Yes
9	Handle outliers	☑ Yes
10	Algorithm selection	☑ Yes

Test Results: 5/5 PASSED ✅

Test	Status	Details
K-Means	✅	Silhouette: 0.61
Hierarchical	✅	Silhouette: 0.58
DBSCAN	✅	Clusters: 3**, Outliers: 7**
Optimal K	✅	K: 3
Comparison	✅	Metrics calculated

Algorithm Performance

K-Means: Silhouette = 0.61**, Inertia = 142.36**
Hierarchical: Silhouette = **0.58, Davies-Bouldin = 0.72**
DBSCAN: Clusters = 3**, Outliers =** 7

Key Findings

- Best algorithm: K-Means
- Reason: K-Means produced the highest silhouette score, well-separated clusters, and consistent grouping for the dataset.
- Clusters found: 3

Conclusion

In this practical, I learned how clustering algorithms group similar data without labels. K-Means worked best for my dataset because it formed clear and well-separated clusters. Hierarchical clustering helped in understanding data structure, and DBSCAN was useful to find outliers. Overall, clustering is helpful for discovering patterns in data.

Status: ✅ Complete | **Next:** Practical 7 - Dimensionality Reduction